Problem Set 5: Behavior Cloning

6.4210/2 Robotic Manipulation — Fall 2026

Due: October 18, 2026

Out: October 8 **Due:** October 18

This problem set is about behavior cloning: learning a policy from demonstrations. The pencil-and-paper problems probe the core difficulties — multimodal demonstrations, the role of observation history, action chunking, and distribution shift — and the computational problem runs a diffusion policy on the Push-T task. Enter the answers in the **Gradescope** blocks in Gradescope; where a question requests a PDF, upload your written response to that question. The **Going further** prompts are for discussion and require no submission. All logarithms are natural.

1 Bookwork

1.1 Behavior cloning

Problem 1 Discrete and continuous action representations [5]

Model the robot as a point in a two-dimensional plane, with horizontal coordinate x and vertical coordinate y. At each discrete step n, it moves one unit to the right. Its scalar action $u[n]$ specifies the vertical coordinate of its next position (in the same units as x and y):

$$x[n+1] = x[n] + 1, \qquad y[n+1] = u[n].$$

There is a diamond-shaped obstacle,

$$|x - 1| + |y| \leq 0.75.$$

We have five demonstrations, each starting at $x[0] = 0$ and stepping through $x = 0, 1, 2$, that all go around the obstacle, some over the top and some under the bottom (Figure 1):

demo	$y[0]$	$y[1]$	$y[2]$	$u[0]$	$u[1]$
a	0	1	0	1	0
b	0.5	1	0	1	0
c	1	1	0	1	0
d	0	−1	0	−1	0
e	−1	−1	0	−1	0

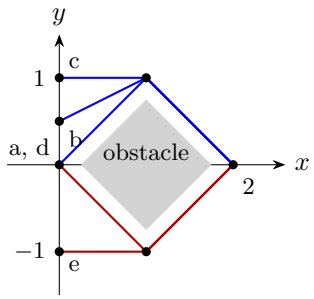


Figure 1: The five demonstrations, over the obstacle (a–c, blue) and under it (d, e, red). Demonstrations a and d both start at the origin.

Gradescope:

1.1 First, fit a simple linear policy, $u = \pi(x, y) = ky$, by least squares on the ten (state, action) pairs in the table. What is k? [1]

$k =$ _____.

1.2 Run the fitted policy from each of the five demonstrations' initial states. How many of the five rollouts hit the obstacle? [1]

Number of rollouts: _____.

1.3 Now suppose we discretize the action, $u \in \{-1, 0, 1\}$, and learn a categorical distribution $p(u \mid x, y)$. Here *RT-style* refers to Robotics Transformer models that discretize continuous robot actions into tokens and predict a categorical distribution over those tokens. We train it by maximum likelihood (equivalently, minimizing the cross-entropy loss) on the five demonstrations. Assume the model is expressive enough to fit the training data as well as possible. What is $p(u \mid x = 0, y = 0)$? Enter $[p(u{=}-1), p(u{=}0), p(u{=}1)]$. [1]

Probabilities: [_____, _____, _____].

1.4 Finally, consider a denoising diffusion policy, which keeps u continuous. At each state, it learns a denoiser $D(\tilde{u}, \sigma)$ that takes a noisy action $\tilde{u} = u + \sigma\epsilon$, with u from the demonstrations and $\epsilon \sim \mathcal{N}(0, 1)$, and predicts the clean u. It is trained by least squares, minimizing $\mathbb{E}\big[(D(u{+}\sigma\epsilon, \sigma){-}u)^2\big]$ over many noise levels σ. At $(x, y) = (0, 0)$, where the demonstrated actions are $u = \pm 1$ with equal probability, the minimizer is [1]

$$D(\tilde{u}, \sigma) = \mathbb{E}[u \mid \tilde{u}] = \tanh(\tilde{u}/\sigma^2).$$

You may use this expression without deriving it. What does $D(0.5, \sigma)$ approach as $\sigma \to \infty$, and as $\sigma \to 0$? Enter $[\lim_{\sigma\to\infty}, \lim_{\sigma\to 0}]$.

Limits: [_____, _____].

1.5 To sample an action, start from a random $\tilde{u}$ and denoise it at a decreasing sequence of noise levels. Use the simplest such sampler, $\tilde{u} \leftarrow D(\tilde{u}, \sigma)$, with $\sigma = 1, 0.5, 0.25$ in turn, starting from $\tilde{u} = 0.2$. What is $\tilde{u}$ after each step? Enter three numbers, to two decimal places. [1]

Values after each step: [_____, _____, _____].

Problem 2 Pauses in the data [6]

Human demonstrators pause: to think, to look, or to reposition their hand. Often times robot learning practitioners try to cut these pauses out of the training data, but that is a slippery slope (what is the sensor threshold for "zero velocity"? and sometimes the policy actually needs to pause... to wait for something to happen in the environment.) Let's see what a pause does to a behavior-cloned policy if we leave it in the training data; it's a good exercise for thinking through some of the implications of longer context lengths and action-sequence prediction.

Consider a robot on a line, $x[n+1] = x[n] + u[n]$, with discrete actions $u \in \{0, 1\}$ (stay or step forward). We have a single demonstration, starting from $x[0] = 0$, that pauses for three steps at $x = 2$:

n	0	1	2	3	4	5	6	7
$x[n]$	0	1	2	2	2	2	3	4
$u[n]$	1	1	0	0	0	1	1	

As in the previous problem, we learn a categorical distribution over actions by maximum likelihood, and assume the model is expressive enough to fit the training data as well as possible.

Gradescope:

2.1 First, a policy with no history, $p(u[n] \mid x[n])$. What is $p(u = 1 \mid x = 2)$? [1]

Probability: _____.

2.2 An RT-style model often decodes greedily, taking the most likely action. OpenVLA, for instance, evaluates its policy with (experiments/robot/openvla_utils.py) [1]

```
action = vla.predict_action(**inputs, unnorm_key=unnorm_key, do_sample=
    False)
```

where `do_sample=False` tells Hugging Face's `generate()` to take the most likely token at each step. Run that greedy policy from $x[0] = 0$. What does $x[n]$ converge to as $n \to \infty$?

Limiting position: _____.

2.3 Now sample from $p(u \mid x)$ instead. The demonstration is at $x = 2$ for four time steps ($n = 2, \ldots, 5$). What is the expected number of time steps the sampled policy is at $x = 2$? [1]

Expected number of steps: _____.

2.4 What is the probability that the sampled policy is still at $x = 2$ at $n = 12$? [1]

Probability: _____.

2.5 Now condition on a history of the last h states, $p(u[n] \mid x[n-h+1], \ldots, x[n])$, so $h = 1$ is the policy above. (Take $x[n] = 0$ for $n < 0$.) What is the smallest h for which no history input in the demonstration appears with two different actions, so that even the greedy policy reproduces the demonstration exactly? [1]

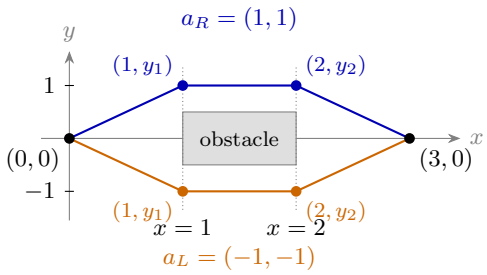
Smallest h: _____.

2.6 Instead, with no history, predict a chunk of K actions, $p(u[n], \ldots, u[n+K-1] \mid x[n])$, execute all K open loop, and then predict again. (Pad the end of the demonstration by repeating its last action, $u = 1$.) With $K = 2$, what is the most likely chunk at $x = 2$, and what is its probability? Enter $[u[n], u[n+1], p]$. [1]

Chunk and probability: [_____, _____, _____].

Problem 3 The average of two good motions [10]

A robot follows straight segments from $(0,0)$ through $(1,y_1)$ and $(2,y_2)$ to $(3,0)$. The x coordinates are fixed, so the policy only predicts $a=(y_1,y_2)$. At the same initial observation o, half the demonstrations use $a_L=(-1,-1)$ and half use $a_R=(1,1)$:



A motion succeeds exactly when both coordinates have the same sign and $|y_1|,|y_2|\geq 1/2$. This simplified rule encodes going around the obstacle on one side. A regression policy minimizes $\mathbb{E}[\|\hat{a}-A\|^2 \mid O=o]$, where A is the demonstrated motion.

Gradescope:

3.1 Give the optimal regression prediction and its expected squared error. $\hat{a}=$ (_____, _____); error: _____. [2]

3.2 A distributional policy samples a_L or a_R, each with probability $1/2$. We score its sample $\hat{a}$ against a demonstration A from the same observation o, where A is a_L or a_R with probability $1/2$ each, independent of the policy's sample. What are the policy's success probability and its expected squared error $\mathbb{E}[\|\hat{a}-A\|^2]$? Success: _____; error: _____. [2]

3.3 Instead, independently sample each waypoint from $\{-1,1\}$ with equal probabilities. What is the success probability? _____. [2]

3.4 A different policy always chooses the complete motion a_L. What is its success probability? _____. More generally, suppose a policy chooses the complete motion a_L with probability p and a_R with probability $1-p$. Which values of p give 100% success? _____. Does 100% success in this task require reproducing the demonstrated 50/50 mix of routes? _____. [2]

3.5 For the policy in the preceding part, minimizing the demonstration negative log likelihood gives $L(p)=-\frac{1}{2}\log p-\frac{1}{2}\log(1-p)$. Find the minimizing $p=$ _____. [2]

Going further: When would reproducing both routes matter, beyond completing this task? How would observing the demonstrator's intended route change the problem?

Problem 4 DAgger: learn from where you actually go [8]

A robot has three nonterminal states: start S, corridor C, and recovery state R. State G is a terminal goal. The actions are F and L. The table gives deterministic transitions, expert labels, and the only two policies we will consider:

State	Next under F	Next under L	Expert	π_A	π_B
S	C	R	F	F	L
C	G	R	F	L	F
R	R	G	L	F	L

The initial dataset contains three labeled examples, $D_0 = [(S, F), (S, F), (C, F)]$: two examples from state S and one from state C. Repeated examples count separately. A rollout starts at state S and stops at state G or after four actions.

Gradescope:

4.1 How many mistakes does each policy make on D_0? π_A: _____; π_B: _____. Which policy makes fewer mistakes? _____. [2]

4.2 Execute π_A from state S for four actions. What are the learner's four actions? _____. List the four states *before* actions and the final state. Visited: _____; final: _____. Give the expert labels at those four visited states: _____. [2]

4.3 One DAgger iteration appends all four (visited state, expert label) pairs from Question 4.2 to D_0, including repeats. How many mistakes does each policy make on this seven-example dataset? π_A: _____; π_B: _____. Which is selected? _____. [2]

4.4 Execute the newly selected policy π_B from state S. List the states along its rollout, including the terminal state: _____. How many actions does the robot take before first reaching goal state G? _____. If instead we added two complete expert trajectories from state S to D_0, would fitting on that expert-only data select a different policy from the original fit? _____. [2]

Going further: How can a policy that disagrees with the expert at state S still complete the task? What would go wrong if we appended the learner's executed actions as labels? We used a pure learner rollout (expert mixing probability $\beta = 0$). Why might early DAgger iterations sometimes execute expert actions too?

Problem 5 Position targets or velocity commands? [9]

A robot hand should stay at position $x = 0$ on a line. We compare predicting a velocity (how fast to move) with predicting a position (where to go), both when obtaining training labels and when executing commands. Use the movement rules below.

Gradescope:

5.1 The hand is stationary, but two measurements taken 0.02 seconds apart report -1 mm and $+1$ mm. Estimate its velocity by dividing the change in measured position by the elapsed time: _____ mm/s. If the same measurement errors occurred 0.01 seconds apart, what velocity would we estimate? _____ mm/s. [2]

5.2 Now suppose position measurements are exact, but the policy makes an error in its prediction. It will not provide a new prediction for one second. During that interval, the robot updates its motion every 0.1 seconds. Compare these two cases, both starting at $x_0 = 0$: [5]

- The velocity command is $+5$ mm/s instead of zero. Each update moves the hand by $0.1(5) = 0.5$ mm: $x_{n+1} = x_n + 0.5$ mm.
- The position target is $b = 1$ mm instead of zero. At each update, the robot measures its current position and moves halfway from there to b: $x_{n+1} = x_n + (b - x_n)/2$.

The errors have been chosen so that both cases make the same first move. Fill in the hand positions, in mm; round to three decimal places.

Update n	Velocity command	Position target
1	_____	_____
2	_____	_____
3	_____	_____
10	_____	_____

If these commands were repeated indefinitely, whose distance from zero would grow without bound: the velocity-commanded hand, the position-commanded hand, or both? _____.

5.3 Finally, suppose both policy predictions are correct: zero velocity and target position zero. A bump displaces the hand to 4 mm just after the prediction. Before the policy can predict again, the robot makes three updates using the same rules as above. Where is the hand after those updates? Velocity command: _____ mm; position target: _____ mm. [2]

Going further: When the velocity policy next observes the displaced hand, could it predict a command that brings the hand back? What advantage does the position target provide while the policy is still processing its next observation? Would the label-noise issue in the first part apply if we recorded velocity commands directly from the teleoperator?

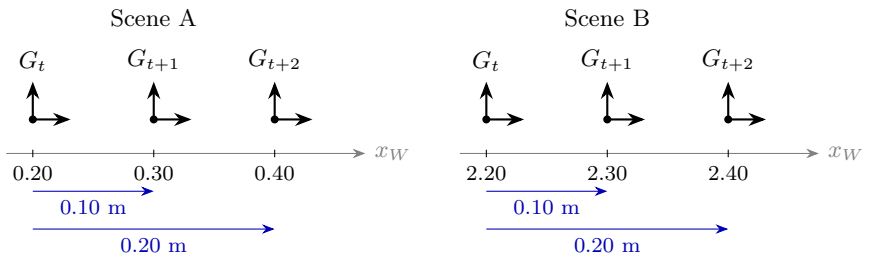
Problem 6 Relative to which frame? [11]

A policy predicts a chunk of two future gripper poses. Use Drake's notation: ${}^{W}X^{G_t}$ is the current gripper pose in the world frame, and ${}^{W}X^{G_{t+j}}$ is a future target pose. All frames in this example have the same orientation, and the hand moves only along the world x axis. Write

$$T(d) = \begin{bmatrix} I_3 & (d,0,0)^{\mathsf{T}} \\ \mathbf{0}^{\mathsf{T}} & 1 \end{bmatrix} ,$$

where d is in meters. Thus ${}^{W}X^{G_t} = T(h)$ places the current hand at world coordinate h.

The desired chunk moves the hand 0.10 m and then 0.20 m forward from its *current* pose. In scene A, $h = 0.20$ m; in scene B, $h = 2.20$ m. The local surroundings move with the hand, so a camera mounted on the hand produces the same image in both scenes.



Both displacements are measured from the same initial frame G_t.

Gradescope:

6.1 Fill in the translation parameters for the two ways of representing this chunk. Each entry below is a pair (d_1, d_2), in meters, representing transforms $(T(d_1), T(d_2))$. [2]

	Absolute: $({}^{W}X^{G_{t+1}}, {}^{W}X^{G_{t+2}})$	Relative: $({}^{G_t}X^{G_{t+1}}, {}^{G_t}X^{G_{t+2}})$
Scene A	_____	_____
Scene B	_____	_____

6.2 First, the policy receives only the camera image; it does *not* receive ${}^{W}X^{G_t}$. Can a deterministic policy always predict the correct absolute chunk in both scenes? _____. Can it always predict the correct relative chunk? _____. The robot's execution code has access to the measured current pose, even when that pose is omitted from the policy's input. [2]

6.3 Now include ${}^{W}X^{G_t}$ in the policy's input. Can either representation express the correct mapping from input to target chunk? _____. Complete the conversion from a relative target to a world target, using transform multiplication: [2]

$${}^{W}X^{G_{t+j}} = _____ .$$

6.4 Even when the current pose is an input, the required output ranges can differ. Suppose the starting position h ranges over $[-10, 10]$ m, and a target's forward displacement d independently ranges over $[0.05, 0.20]$ m. The absolute translation is $h + d$; the relative translation is d. Give their ranges: absolute _____ m; relative _____ m. [5]

Suppose each scalar output is linearly scaled to $[0, 1]$ during training. An error of 0.01 in that scaled output corresponds to what translation error after undoing the scaling? Absolute: _____ mm; relative: _____ mm. Which representation needs a more accurate scaled prediction to meet a 1 mm position tolerance? _____.

Going further: Does supplying the current pose make the two representations equally easy to learn? Does the range calculation prove that absolute targets must be harder? How could a network predict a small displacement but still provide a world-frame target to the robot?

Problem 7 Sharing and reweighting demonstrations [8]

We want to train a policy for a seven-degree-of-freedom robot, A. We have demonstrations from A, another type of seven-degree-of-freedom robot, B, and a six-degree-of-freedom robot, C. Their joint axes, link lengths, and joint limits differ. Assume the observations have already been put in a common format, and that gripper poses use consistent coordinate frames and the same definition of the gripper's tool frame.

Gradescope:

7.1 Suppose the policy predicts joint angles for robot A. Without converting the labels, can B's seven joint angles be used as target joint angles for A? _____. Can C's six joint angles be used? _____. [2]

7.2 Suppose instead that the policy predicts gripper poses ${}^{W}X^{G}$. Can demonstrations from all three robots use this same action representation? _____. Does a gripper pose reached by B or C have to be reachable by A? _____. [2]

7.3 We use gripper-pose labels and decide to give more influence to data from A. Let $q(r)$ be the fraction of examples from robot r and $p(r)$ the desired fraction of total training weight: [2]

Robot	A (our robot, 7 DOF)	B (other 7 DOF)	C (6 DOF)
$q(r)$	0.2	0.3	0.5
$p(r)$	0.6	0.3	0.1

These desired fractions are our choice, not something determined by the number of joints. Compute the importance weights $w(r) = p(r)/q(r)$: $w(A) =$ _____, $w(B) =$ _____, $w(C) =$ _____. In a batch of ten examples with two from A, three from B, and five from C, what fraction of the total weight belongs to A? _____.

7.4 A fixed policy has average prediction losses 0.1, 0.2, and 0.4 on examples from A, B, and C, respectively. Compute the unweighted average loss under q: _____. Compute the weighted average $\sum_r q(r)w(r)\ell(r)$: _____. [2]

Going further: Even after expressing every action as a gripper pose, why might data from B and C be less useful than data from A? Think about the sets of hand poses and paths each robot can execute. Can importance weighting make an infeasible target feasible?

Problem 8 When should the robot ask for another demonstration? [5]

Two independently trained ensemble members predict a probability of taking the left route, $p_1(L \mid o)$ and $p_2(L \mid o)$. Define their average left-route probability $\bar{p}$ and their disagreement U as

$$\bar{p} = \frac{p_1 + p_2}{2}, \qquad U = \frac{(p_1 - \bar{p})^2 + (p_2 - \bar{p})^2}{2}.$$

We also compute $d(o)$, the distance to the nearest training observation in a fixed learned feature space. All distances below use the same scale.

Observation	$p_1(L \mid o)$	$p_2(L \mid o)$	$d(o)$
o_A	0.5	0.5	0.1
o_B	0.1	0.9	0.2
o_C	0.0	0.0	2.0

Gradescope:

8.1 Give $(\bar{p}, U)$ for each observation. [3]

o_A: (_____, _____); o_B: (_____, _____); o_C: (_____, _____).

8.2 Request an expert label whenever $U > 0.10$ *or* $d(o) > 1.0$. Which observations trigger a request? [2]
_____. Which would be missed by using only the disagreement threshold? _____.

Going further: At both o_A and o_B, averaging the two models' predictions gives probability 1/2 for each route. Why might asking for more data still be more useful at o_B? Can a unanimous ensemble still be wrong? Is either U or feature distance automatically a calibrated failure probability? How could queried labels feed back into the DAgger procedure?

Problem 9 Flow matching between noise and two data points [12]

Flow matching generates data by transporting samples from a simple noise distribution to the data distribution. Let $X_0 \sim \mathcal{N}(0, 1)$, and independently choose X_1 with equal probability from the two data points $x^{(1)} = 2$ and $x^{(2)} = 4$. Define a straight path between them:

$$X_t = (1 - t)X_0 + tX_1 \ , \qquad 0 \le t \le 1 \ .$$

Here noise is at $t = 0$ and clean data at $t = 1$. Use $t < 1$ when writing densities or velocities. You may write $\mathcal{N}(x; \mu, \sigma^2)$ for a Gaussian density evaluated at x; there is no need to expand its exponential.

Gradescope:

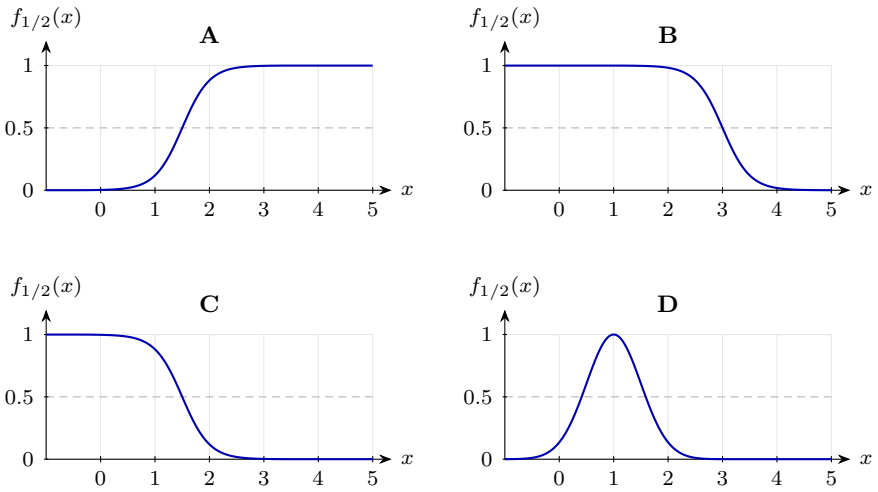
9.1 Fix $X_1 = x_1$ and t. What is the conditional distribution of X_t? Specify its mean and variance as functions of t and x_1. [2]

Mean: _____; variance: _____.

9.2 Now average over the two equally likely choices of X_1. Write the resulting density $p_t(x)$ of X_t. [2]

9.3 Observing $X_t = x$ gives information about which endpoint was chosen. Write the posterior probability $f_t(x) = \Pr(X_1 = 2 \mid X_t = x)$ in terms of the Gaussian densities. Upload your formula and a brief derivation as a PDF response to Question 9.3 in Gradescope. [2]

9.4 Which sketch shows $f_{1/2}(x)$, the probability that the clean endpoint is 2, given the noisy value x at time $t = 1/2$? Choose A, B, C, or D: _____. [2]



Setup for Questions 9.5 and 9.6. For a fixed endpoint x_1, the velocity field

$$v_t(x \mid x_1) = \frac{x_1 - x}{1 - t}$$

transports a point along its straight path to x_1 by time 1. To transport the noise distribution to a mixture of endpoints, use the conditional average

$$v_t(x) = \mathbb{E}\left[\frac{X_1 - x}{1 - t} \,\middle|\, X_t = x\right] \quad .$$

This velocity averages over the possible endpoints given the current noisy value, rather than choosing one endpoint in advance.

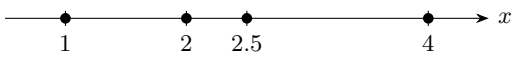
Gradescope:

9.5 Express $v_t(x)$ using $f_t(x)$ and the two endpoint velocities. Upload your formula and a brief explanation of the posterior weights as a PDF response to Question 9.5 in Gradescope. [2]

9.6 At $t = 1/2$, indicate the direction of the velocity at the four marked positions in the figure. Give left or right at $x = 1, 2, 2.5, 4$, respectively: [2]

_____, _____, _____, _____.

Evaluate $v_{1/2}(x)$ at the four marked positions.



Going further: For a fixed x away from 2, 3, and 4, describe the direction and magnitude of $v_t(x)$ as t approaches 1.

Problem 10 How much evidence is enough? [8]

Two robot policies, A and B, are evaluated on a task with a binary outcome: success or failure. Each trial starts from an independently sampled configuration from the same distribution, and the testing conditions are the same for both policies. Let p_A and p_B be their true success probabilities.

Consider three separate experiments, each with N trials per policy:

N	Policy A: successes/trials	Policy B: successes/trials
5	4/5	3/5
10	8/10	6/10
100	80/100	60/100

In each case, the sample size and a *two-sided* test of $H_0 : p_A = p_B$ against $H_1 : p_A \neq p_B$ were chosen in advance. Use significance level $\alpha = 0.05$ for each experiment considered on its own. Round numerical answers to three significant digits.

Gradescope:

10.1 Use Fisher's exact test to compute the p-value for each experiment. Is the difference statistically significant at the specified level? You may use statistical software: the input table has rows for A and B and columns for successes and failures. [3]

N	Exact p-value	Significant? (yes/no)
5	_____	_____
10	_____	_____
100	_____	_____

10.2 Now use the normal approximation. With observed rates $\hat{p}_A = s_A/N$, $\hat{p}_B = s_B/N$, and pooled rate $\hat{p} = (s_A + s_B)/(2N)$, compute [3]

$$z_{\text{obs}} = \frac{\hat{p}_A - \hat{p}_B}{\sqrt{2\hat{p}(1-\hat{p})/N}} \quad , \qquad p_{\text{approx}} = 2\big[1 - \Phi(|z_{\text{obs}}|)\big] \quad ,$$

where Φ is the standard normal cumulative distribution function. Also check whether each group has at least 5 expected successes and 5 expected failures under the null: $N\hat{p} \geq 5$ and $N(1-\hat{p}) \geq 5$.

N	z_{obs}	Approximate p-value	Passes count check?
5	_____	_____	_____
10	_____	_____	_____
100	_____	_____	_____

10.3 What is A's observed advantage over B in all three experiments? _____ percentage points. Do the first two experiments establish that $p_A = p_B$? _____. [2]

Going further: Why does increasing N change the strength of the evidence even though the observed success rates stay the same? Does statistical significance alone tell us whether the improvement matters for the robot's application?

Problem 11 Representing 3D rotations [12]

A robot policy might receive the hand's orientation as an input or predict a desired orientation as an output. Keep the coordinate frames fixed, and write R for the rotation matrix ${}^{W}R^{G}$. We will compare several ways to encode this rotation as numbers. Recall that unit quaternions q and $-q$ represent the same rotation.

Consider three possible difficulties:

J: nearby rotations, distant numbers.
Rotations arbitrarily close to one another can have numeric representations far apart in ordinary Euclidean distance. This concerns the inputs or training labels, not whether the formula converting numbers to a rotation is continuous.

U: non-unique.
Two different valid numeric representations can describe exactly the same rotation.

C: constrained.
Not every real-valued tuple is already a valid instance of the stated representation; conditions such as non-negativity or orthogonality must be enforced.

Classify the representations *as stated*, before any repair of network outputs.

Gradescope:

11.1 For each row, list all applicable letters from $\{J, U, C\}$, or write "none." No angle wrapping or quaternion sign alignment is assumed unless specified in the row. [5]

Representation	Difficulties
Roll–pitch–yaw angles $(\phi, \theta, \psi) \in \mathbb{R}^3$, with $R = R_z(\psi)R_y(\theta)R_x(\phi)$ and no restrictions on angle ranges.	_____
Unit quaternion $q \in \mathbb{R}^4$, allowing both signs.	_____
Unit quaternion with its sign chosen so that its first nonzero component is positive (components ordered scalar first, then x, y, z).	_____
All nine entries of R.	_____
The second and third columns of R, stored as six numbers. These are actual matrix columns, not arbitrary vectors that will later be converted into columns.	_____

11.2 Imagine that we trained a network to predict the second and third columns of R. At execution time, it outputs $u, v \in \mathbb{R}^3$, which need not be unit length or orthogonal. How can we turn them into a valid rotation? Use u to fix the direction of the second column r_2, then remove from v its component along r_2. Assume $u \neq 0$ and v is not parallel to u. Fill in the normalization, projection, and cross-product steps: [4]

$r_2 =$ _____; $w =$ _____.

$r_3 =$ _____; $r_1 =$ _____.

Assemble the result as $R = [\, r_1 \; r_2 \; r_3 \,]$.

11.3 Apply your construction to $u = (2, 0, 0)^{\mathsf{T}}$ and $v = (1, 0, 3)^{\mathsf{T}}$. Give the resulting first column: $r_1 =$ _____. What is $\det R$? _____. If we reversed the cross-product order in the last step, what would the determinant be? _____. [3]

Going further: What goes wrong when $u = 0$ or when u, v are parallel? After the conversion, can different raw network outputs produce the same rotation? Which difficulties matter when these representations are used as inputs, and which require action when they are predicted as outputs?

Background reading. Ross, Gordon, and Bagnell's DAgger paper; and Zhou et al., *On the Continuity of Rotation Representations in Neural Networks*. Problem 9 is adapted from Question 6 of the 6.7900 Fall 2024 alternative final exam.

1.2 Additional problems for 6.4212

Problem 12 Diffusion: adding noise and learning to remove it [8]

Consider a diffusion model in one dimension. Its forward process is

$$X_t = \sqrt{1-\beta_t}\, X_{t-1} + \sqrt{\beta_t}\, Z_t \ , \qquad Z_t \sim \mathcal{N}(0,1) \ ,$$

where each Z_t is fresh independent noise. Define $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. At a chosen noise level, we can sample directly from the same conditional distribution using

$$X_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, Z \ , \qquad Z \sim \mathcal{N}(0,1) \ .$$

Here x_0 is a fixed clean example. The new Z represents the combined noise in this direct construction, rather than a particular step's Z_t. Write $q_t(x_t \mid x_0)$ for the resulting conditional density.

Gradescope:

12.1 Give the mean and variance of $q_t(x_t \mid x_0)$. [2]

Mean: _____; variance: _____.

12.2 Suppose $\bar{\alpha}_T = 0.8$, and we start the reverse sampler from a standard normal. Does this choice of T ensure that the forward process has become close to standard Gaussian noise? _____. [1]

12.3 One training update uses a sampled noise value z in [5]

$$\nabla_\theta \left| z - \epsilon_\theta\big(\sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, z, t\big)\right|^2 \ .$$

For each quantity below, identify its expression in this update, or write "not present." You may abbreviate the network's first input as x_t.

(i) A sample from $q_t(x_t \mid x_0)$: _____.

(ii) The mean of $q_t(x_t \mid x_0)$: _____.

(iii) The standard deviation of $q_t(x_t \mid x_0)$: _____.

(iv) A prediction of the individual forward-step noise Z_t: _____.

(v) A prediction of the standardized combined noise Z used in the direct construction from x_0 to x_t: _____.

Going further: Given the current value x_t, a reverse sampler may draw fresh noise $Z \sim \mathcal{N}(0, 1)$ and set

$$X_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t Z \quad .$$

What changes if we simply set $\sigma_t = 0$ at every step? Must every sample then be identical? Must the output distribution remain correct?
Suppose we instead try to generate a clean example in one step using $\hat{x}_0 = x_T - \epsilon_\theta(x_T, T)$. What is wrong with this expression? Even after correcting its scaling, why can a single denoising prediction have difficulty producing a good sample when $\bar{\alpha}_T$ is very small?

2 Computational questions

2.1 Hands on with Diffusion Policy

Problem 13 Diffusion Policy on Push-T [8]

For this problem, you'll experiment with a slightly modified "Push-T" example from the original Diffusion Policy paper, develop your intuition, and answer some basic questions.

You'll work in `diffusion_policy_pusht_state.ipynb`. It includes the code for the Push-T environment and diffusion policy, so no separate Python files are needed. The notebook downloads a pretrained policy, and should run locally without GPU resources.

Setup. Open the notebook in Colab or Jupyter and run its setup cells in order. They install the dependencies and load the bundled helpers; the helper implementation is hidden in a collapsed setup cell. First, just step through the notebook and make sure you understand each of the steps.

The `show_action_samples` method is useful for visualizing the learned distribution. For each of the initial conditions below, indicate whether the distribution is uni-modal or bi-modal and whether the motions look appropriate.

- Keeping the T at its initial position.
  - Compare the action distribution for pusher positions (relative to T) $0, -100$ vs $100, -100$.
  - Compare the action distribution for pusher positions (relative to T) $100, 50$ vs $50, 50$.
  - Compare the action distribution for pusher positions (relative to T) $0, -100$ vs $0, -300$.
  - Compare the action distribution for pusher positions (relative to T) $0, -100$ but now initialize the velocity to $100, 0$ and then $-100, 0$.
- Displace the T 100 pixels in the +x direction. Repeat the pusher displacement tests.